Now test yourself

TESTED

1 State what is meant by the term 'rotary movement'. (1)
2 Give an example of a class 2 lever. (1)
3 Name **three** types of follower in a cam and follower mechanism. (3)
4 Describe the function of a crank and slider mechanism. (2)
5 Describe the purpose of an idler gear. (2)

1.6 Electronic systems

1.6.1 Sensors

REVISED

- Sensors are input devices. They take signals from the environment around them, such as light and temperature levels, and turn them into electronic signals, such as current, voltage or resistance.
- **LDRs** change the light level that is detected into a resistance. The resistance decreases as the brightness increases. LDRs are often used in outdoor street lamps and night lighting.
- **Thermistors** change the temperature level that is detected into a resistance. Thermistors are often used where it is important to know the temperature, such as inside a greenhouse or a refrigerator.

Thermistor: a sensor input that can detect changes in temperature levels.



Figure 1.17 A light-dependent resistor (LDR)



Figure 1.18 A push to make switch

1.6.2 Control devices and components

REVISED

- **Switches** are used to either 'make' (allow current to flow through) or 'break' (not allow current to flow through) a circuit. An example is a light switch that turns a lamp 'on'.
- Switches are described by either the number of poles and throws that they have, or by their function. For example, a push to make switch 'makes' a circuit when it is pressed.
- Other examples of switches include rocker switches, toggle switches, tilt switches, reed (magnetic) switches and micro switches.
- **Transistors** are used as either an electronic switch or a current amplifier. A transistor has three leads called the base, collector and emitter. When the voltage at the base reaches approximately 0.6 V, current is allowed to flow between the collector and the emitter.
- **Resistors** are used to reduce the flow of current in a circuit. The higher the value of a resistor, the more resistance it has. The amount of resistance is measured in ohms, or Ω.

Switch: a component that can be used to either 'make' or 'break' a circuit.

Transistor: a component that acts as either an electronic switch or a current amplifier.

Resistor: a component that reduces the flow of current.

1.6.3 Outputs

REVISED

- Outputs take electronic signals and turn them back into 'real world' signals, such as light or sound.
- **Buzzers** use an internal oscillator to produce sounds at different frequencies when current flows through them. They are often used in alarms, warning systems and electronic children's toys.

Buzzer: an output device that produces sound when current flows through it.

- **LEDs** produce light when current flows from the anode to the cathode. They can be damaged by too much electrical current, so they usually require a protective resistor connecting in series with them.
- LEDs use far less current and thus save energy. They also last much longer as they do not 'burn out'.

Typical mistake

Misunderstanding the difference between input, control and output components and their main functions in a system.

Now test yourself

TESTED

1 A designer is developing an idea for an outdoor security light. The light must come on automatically when it is dark. Name a suitable input sensor and output device for this system. (2)
2 Describe how a transistor operates when being used as an electronic switch. (3)
3 State the main function of a resistor in a circuit. (1)
4 Give **two** applications of buzzers in electronic systems. (2)

Exam tip

Make sure you understand the working characteristics, applications, advantages and disadvantages of the main input, control and output components.

1.7 The use of programmable components

1.7.1 How to make use of flowcharts

REVISED

- Programmable components are used to replace analogue integrated circuits. They can be programmed to perform tasks such as responding to sensors, producing time delays and counting.
- An example is a microcontroller, which has several pins for the connection of different input and output devices.
- Flowchart software is commonly used to program microcontrollers. Standard symbols are joined together using arrows to create a set of instructions for how the system will operate. This is then downloaded on to the chip. Each symbol represents a type of command that can be edited for a specific purpose within the program.
- This type of programming is highly visual, so it is easier to make changes or spot errors. Complex programming knowledge is not necessary. Programs can be simulated on-screen to check that they will work.

Typical mistake

Not using the correct flowchart symbols for each programming command – make sure you know when each of these should be used in a program.

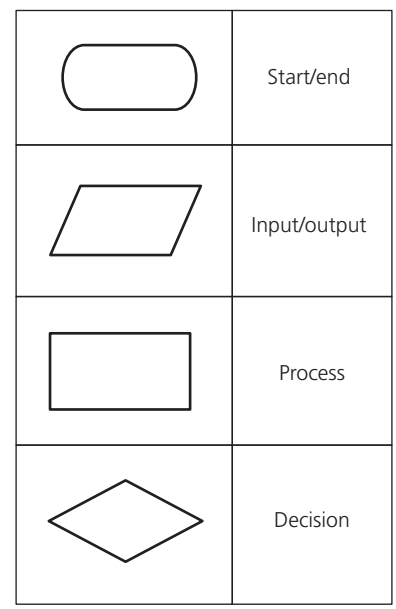


Figure 1.19 Common flowchart symbols

1.7.2 How to switch outputs on/off

REVISED

- In most flowchart software, an output is turned on by inserting the word 'high', followed by the pin number that the output is connected to, into the **input/output** command symbol.
- For example, 'High 2' would turn on the output connected to pin 2. To turn the output off, the word 'low' is used instead.
- **Decision** commands are used to check the state of a digital input device, such as a switch. This is achieved by inserting a question into

the decision command symbol, for example 'is pin 3 on?'. If the answer is yes, the program follows the 'yes' (Y) arrow; if it is no, it follows the 'no' (N) arrow.



Figure 1.20 A command that would turn an output device connected to pin 2 'on'

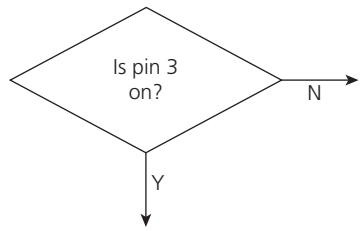


Figure 1.21 A decision command checking whether a digital input device connected to pin 3 is 'on'

Start/End: a command that marks the start or end of a program.

Input/output: a command that is used to change the state of an input or output pin.

Decision: a command that checks the state of a digital input device.

1.7.3 How to process and respond to analogue inputs

REVISED

Many sensors, such as LDRs and thermistors, produce analogue signals. A special type of decision command, called **compare**, can be used to check whether an analogue signal is greater or less than a particular value, known as the threshold.

Compare: a command that checks whether an analogue value is greater or less than the threshold.

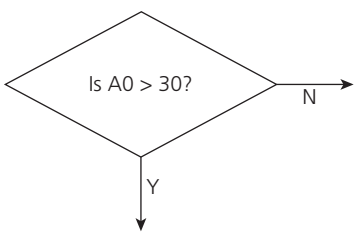


Figure 1.22 A compare command checking whether an analogue input signal is greater than 30

1.7.4 How to use simple routines to control outputs with delays, loops and counts

REVISED

- One of the main reasons for using programmable components is that complex **process** commands can be created with less circuitry, as programming replaces the hardware.
- To create time delays the **wait** command can be used. For example, the program in Figure 1.23 will turn on an output connected to pin 2 when pin 3 detects that a switch has been pressed. The output will remain on for one second and then turn off. The program will then wait another second before looping back to repeat, creating a continuous pulsing sequence.
- Most flowchart programming software has a type of process command that can **count** the number of times an input pin goes from low to high within a certain time period, for example how many times a switch is pressed.

Process: a command that can be used to perform processing functions such as counting or creating time delays.

Wait: a command that is used to create time delays.

Count: a command that can count the number of times an input pin goes from low to high.

Assembly drawings

Assembly drawings show how all the different parts of a product fit together. Each part is identified by a number. A bill of materials is included that lists each part needed and the quantities required.

Assembly drawing: drawing that shows how all the different parts of a product fit together, with each part identified by a number.

Schematic diagram: uses symbols to show a circuit in terms of its individual components and how they are connected.

System and schematic diagrams

- System, or block, diagrams show a system in terms of its input, processes and output subsystems. This presents a top-down overview of the system.
- **Schematic diagrams** show a circuit in terms of the individual components used. These are represented using symbols; the connections between them are shown with straight lines.

Computer-aided design

CAD software can be used to produce both drawings and virtual models of designs. Using CAD has many advantages, such as:

- Accuracy of drawing and modelling is very high.
- Changes can be made quickly and easily.
- No materials or physical resources are needed, reducing costs.
- Models can be simulated on screen.
- 2D drawings can be produced directly from the 3D model, saving time.
- CAD files can be outputted directly to CAM equipment for manufacture.

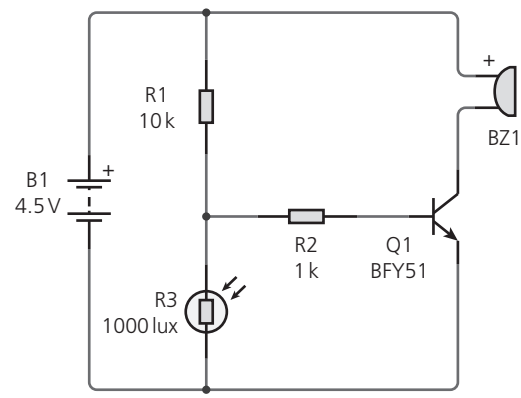


Figure 1.37 An electronic transistor circuit schematic

Exam tip

Make sure you know the purpose and correct use of each of the various ways of developing and communicating design ideas.

1.17.2 Recording and justifying design ideas clearly and effectively

REVISED ☐

- Designers must be able to justify their ideas and the design decisions that they have taken, for example giving reasons why certain materials and components have been chosen. This should always be in terms of the needs and wants of the end user and how well each design would meet them.
- This can be achieved through detailed annotations of design ideas, or listing how each one meets the different requirements of the design specification. Consideration should be given to the wider design context and how each idea fits within this.

Now test yourself

TESTED ☐

1 Describe the differences between freehand sketching and formal drawing. (4)
2 Name **two** types of perspective drawings. (2)
3 Explain the differences between first and third angle orthographic projections. (4)
4 Describe the purpose of a systems diagram. (2)
5 Give **two** reasons why CAD is used to produce drawings of products. (2)